Eugene Lee

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Summary

AI Researcher specializing in efficient deep learning, vision-language models, and in-context learning for LLM. Expertise in model compression, resource-constrained architectures, and multimodal alignment with focus on reducing inference costs and maximizing data efficiency. Proven track record of translating research into production-ready solutions with real-world impact.

Skills: PyTorch, **Python, C/C++**, CUDA, TypeScript/JavaScript, MATLAB, Verilog, AWS, Docker, iOS, Android, SQL, Redis

Education

National Yang Ming Chiao Tung University, Hsinchu *September 2017 - July 2023*
Ph.D. in Electronics Engineering, Advisor: Prof. Chen-Yi Lee; GPA: 4.0/4.0
National Chiao Tung University, Hsinchu *August 2013 - June 2017*
B.S. in Electronics Engineering

Selected Publications

h-index: 6; Citations: 429; Total: 12

Eugene Lee and Chen-Yi Lee, "**NeuralScale: Efficient Scaling of Neurons for Resource-Constrained Deep Neural Networks**," In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), June 2020 (**Oral**)

Eugene Lee, Evan Chen and Chen-Yi Lee, "**Meta-rPPG: Remote Heart Rate Estimation Using a Transductive Meta-Learner**," In Proceedings of the European Conference on Computer Vision (**ECCV**), August 2020

Eugene Lee, Cheng-Han Huang and Chen-Yi Lee, "**Few-Shot and Continual Learning with Attentive Independent Mechanisms**," In Proceedings of the International Conference on Computer Vision (**ICCV**), October 2021

Eugene Lee, Lien-Feng Hsu, Evan Chen and Chen-Yi Lee, "**Cross-Resolution Flow Propagation for Foveated Video Super-Resolution**," In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), January 2023

Eugene Lee, Annie Ho, Yi-Ting Wang, Cheng-Han Huang and Chen-Yi Lee, "**Cross-Domain Adaptation for Biometric Identification Using Photoplethysmogram**," International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), May 2020

Chen, Rui, Eugene Lee, Yuxin Wang, Aditya Yadav, Minling Zhong, Pragti, Yujie Sun, and Jiajie Diao. **A Versatile Near-Infrared Fluorescent Probe for Fast Assessment of Lysosomal Status via a Large Multimodal Model.**, Aggregate: e70118, 2025.

Yadav, Aditya, Eugene Lee, Rui Chen, Soryong R. Chae, Yujie Sun, and Jiajie Diao. **Understanding the Role of Polyethylene Glycol Coating in Reducing the Subcellular Toxicity of MXene Nanoparticles Using a Large Multimodal Model.**, Materials Today Bio (2025): 102372.

Eugene Lee and Chen-Yi Lee, **PPG-Based Smart Wearable Device with Energy-Efficient Computing for Mobile Health-care Applications**, IEEE Sensors Journal, vol. 21, no. 12, pp. 13564-13573, 15 June, 2021, doi: 10.1109/JSEN.2021.3069460.

Patents, Awards, and Services

Patents: Video display systems (US 12,211,174, 2025); Physiological sensing method (US App 16/441,801).

Awards: Founder University Cohort 11 (2025); Future Tech Awards (2024); Draper University Entrepreneurship (2022); Novatek Ph.D. Fellowship (2020-2022); Broadcom Foundation Scholarship (2019); Synopsys ARC Contest (3rd Place, 2017).

Academic Service:

- **Reviewer:** CVPR ('22 ∼ '26), ICCV ('21, '25), ECCV ('22, '24), NeurIPS ('25), WACV ('26).
- **Invited Talks:** "Semiconductor Chips for Fast Medical Testing" (Taiwan-Malaysia Forum '24); "Going Beyond Neural Architecture Search" (Academia Sinica '20); "NeuralScale" (NTU '20).

Work Experience

University of Cincinnati, Ohio — August 2024 – Present
Postdoctoral Researcher, College of Medicine

- Developed novel **in-context learning** framework using Large Language Models to assess drug effects through **few-shot demonstrations**, achieving competitive performance while reducing labeled data requirements by 10x.
- Researched **demonstration selection strategies** that improve model reasoning quality and reduce context window size by 40%, directly addressing efficiency challenges in LLM inference.
- Built optimized imaging pipelines with temporal modeling for biological workflows, demonstrating expertise in **efficient multi-modal system design**.

Paidge, Inc. — January 2024 – Present
Founder

- Created an object-intelligence platform combining vision encoders with LLMs for conversational object interaction, deploying **efficient Vision-Language Models (VLMs)** using vLLM for production.
- Designed proprietary multi-frame aggregation model with FAISS for **large-scale instance retrieval**, optimizing embedding quality while maintaining sub-100ms query latency.
- Re-engineered lightweight GraphRAG for **high-throughput indexing**, reducing context requirements through structured knowledge representation and achieving 3x faster inference.
- Conducted pilot deployments generating 10+ actionable insights, demonstrating 40% improvement in visitor engagement over traditional audio guides.

National Yang Ming Chiao Tung University, Hsinchu — August 2023 – July 2024
Postdoctoral Researcher, Institute of Electronics

- Directed architectural study optimizing **Vision-Language Pipelines** to improve token efficiency, reducing visual token count by 60% through **adaptive token reduction** while maintaining task performance.
- Researched **efficient attention mechanisms** for VLMs that minimize computational overhead in long-context scenarios, achieving 2x speedup in end-to-end latency.
- Led design of BioFPGA platform integrating CMOS ASIC with embedded systems, demonstrating hardware-software co-design expertise.

Advanced Bio Chips, Hsinchu — August 2023 – July 2024
CEO

- Co-led BioFPGA commercialization, securing approximately $1M in funding and driving cross-functional integration across IC design, embedded software, and bioassay teams.
- Delivered system architecture optimizations that improved inference throughput by 5x while meeting strict power and memory constraints for edge deployment.